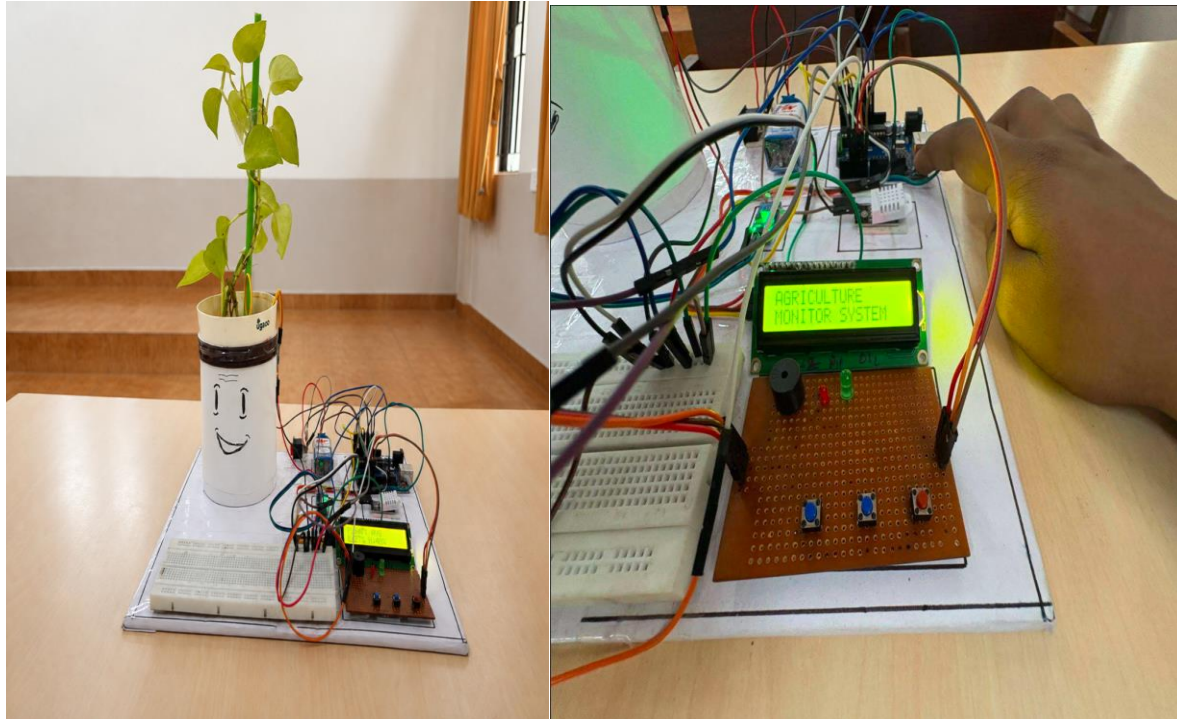


Agriculture Monitoring System

Arduino UNO + ESP32-S3 Based Smart Agriculture Monitoring Project

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Introduction

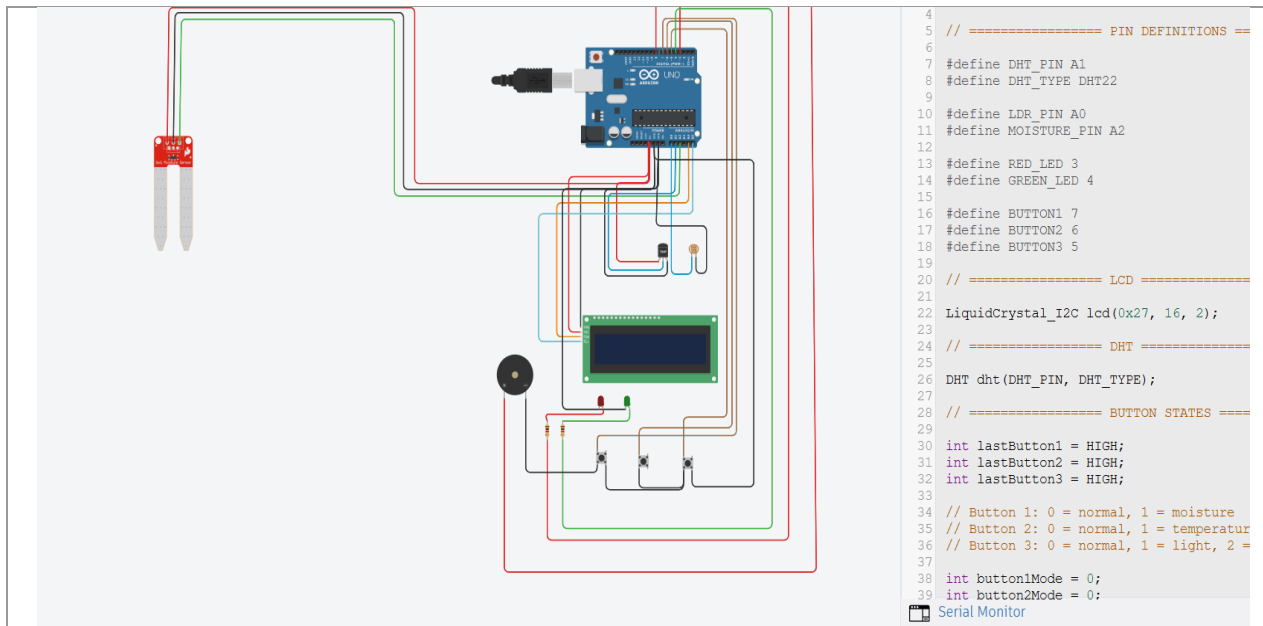
This project is a simple agriculture monitoring system built using an Arduino UNO and an ESP32-S3 microcontroller. The system reads temperature, humidity, light intensity and soil moisture from sensors connected to the Arduino UNO, and uses this data to figure out the overall condition of the plant. The Arduino UNO shows the readings on an LCD and also sends the data to the ESP32-S3 through UART, which then connects to Wi-Fi. The goal of this project was to build an affordable setup that can give a quick idea of how healthy the growing conditions are, without needing to check everything manually.

Components Used

- Arduino UNO
- ESP32-S3
- DHT22 Temperature and Humidity Sensor
- LDR (Light Dependent Resistor)
- Soil Moisture Sensor
- 16x2 I2C LCD Display
- Red LED (temperature alert)
- Green LED (good condition indicator)
- 3 Push Buttons (for switching LCD display modes)
- Jumper wires, breadboard and resistors as needed

Circuit Diagram

The circuit connects the DHT22, LDR and soil moisture sensor to the analog and digital pins of the Arduino UNO, along with the I2C LCD, the two LEDs and the three push buttons. The Arduino UNO's TX pin is connected to the ESP32-S3's RX pin through a voltage divider, since the UNO works on 5V logic while the ESP32-S3 works on 3.3V logic.



Working

The Arduino UNO continuously reads the sensor values. The LDR and soil moisture readings are converted into percentages, while the DHT22 gives direct temperature and humidity values. Based on set threshold ranges, the Arduino decides whether the plant condition is GOOD, AVG or BAD, and also estimates approximate NPK values based on this condition.

The three push buttons let the user switch between different values on the LCD, such as soil moisture, temperature, humidity, light intensity, plant condition and the NPK values. The red LED turns on when the temperature goes above 30°C, and the green LED turns on when the plant condition is GOOD and there is no temperature alert.

At the same time, the Arduino UNO sends all this data to the ESP32-S3 over UART. The ESP32-S3 connects to the Wi-Fi network and prints the received data along with its IP address on the Serial Monitor, which can be used later to build a web dashboard or app on top of this system.

Future Scope

- Web dashboard to view live sensor data
- Automatic irrigation based on soil moisture
- Real NPK sensor for accurate nutrient readings
- Data logging and remote alerts

Conclusion

This project combines simple sensor readings, an LCD display, LED indicators and basic Wi-Fi connectivity to create a low-cost agriculture monitoring setup. It gives a good starting point for anyone who wants to build a smarter, IoT-based version of this system in the future.